

AI + Climate Citation Prediction

Full Process and Research Methodology

Based on the submitted project presentation and supplied PySpark notebook

Research question: Can publication metadata available around publication time identify AI + climate papers that will become highly cited, and does adding an early citation signal improve prediction?

1. Project Overview

Problem

Predict citation impact for papers at the intersection of artificial intelligence and climate research, where future citations are unknown but publication metadata can be available early.

Objective

Classify papers into high-impact and non-high-impact groups using a deterministic top-10% citation threshold within each publication-year cohort.

Models

Logistic Regression, Random Forest, and Gradient-Boosted Trees (GBT).

2. End-to-End Process

Pipeline

Define scope → retrieve OpenAlex records → clean data → construct yearly top-10% target → engineer metadata features → optionally add early citation signal → chronological split → fit preprocessing on training data → train models → tune threshold on validation data → evaluate final test → compare → interpret → export.

Leakage control

The 2023–2024 test set is not used for threshold selection or model-selection decisions.

3. Data Collection

Source

OpenAlex supplies bibliographic records and fields for dates, citations, authors, institutions, countries, venues, open-access status, topics, and yearly citation information.

Retrieval

Records are collected page-by-page through the API until the configured limit is reached or pagination ends.

Robustness

The cleaned workflow includes retry handling for temporary API failures and supports reuse of a saved CSV to avoid repeated downloads.

4. Data Preparation

Unit

One row represents one research work/paper.

Cleaning

Normalize publication year and citation fields, remove unusable records, and transform structured metadata into modeling variables.

Target source

Total citations are used to construct the target, not as a predictor in the primary metadata-only experiment.

5. Target Construction

Definition

high_impact_label = 1 if the paper is in the top 10% of citations for its publication year; otherwise 0.

Reason

Year-specific thresholds reduce the advantage older papers have from having more time to accumulate citations.

Imbalance

The positive class is approximately 10% by design, so Logistic Regression uses class weighting and tree models use minority oversampling.

6. Feature Engineering

Numeric metadata

Number of authors, number of institutions, number of countries, international-collaboration indicator, authors per institution, open-access indicator, and number of topics.

Categorical metadata

Venue and open-access status are encoded through the Spark ML preprocessing pipeline.

Primary setting

Metadata-only; citation counts are excluded from predictors.

Secondary setting

Adds early_calendar_year_citations. This is a calendar-year signal, not an exact first-12-month citation measure.

7. Temporal Experimental Design

Train

2020–2021.

Validation

2022; used for development and threshold selection.

Test

2023–2024; untouched until final evaluation.

Rationale

Chronological splitting better represents deployment on future publications and reduces temporal leakage.

8. Spark ML Preprocessing

Pipeline

Numeric and categorical transformations produce the Spark features vector consumed by the classifiers.

Fit discipline

Preprocessing is fitted using training data and then applied to validation/test data.

9. Model Training

Logistic Regression

Linear baseline with class weighting.

Random Forest

Nonlinear tree ensemble with configured tree count/depth and sqrt feature selection; trained on balanced training data.

GBT

Gradient-boosted tree ensemble with configured iterations/depth; trained on balanced training data.

Comparison

Three models × two feature settings = six experimental combinations.

10. Evaluation

ROC-AUC

Measures ranking quality across classification thresholds.

PR-AUC

Measures precision-recall behavior and is particularly important for the approximately 10% positive class.

Classification metrics

Precision, recall, F1, accuracy, and TP/FP/FN/TN are reported.

Interpretation

Accuracy is secondary because class imbalance can make it misleading.

11. Threshold Selection

Need

A 0.5 probability threshold is not necessarily optimal for an imbalanced classification task.

Search

Evaluate thresholds from 0.10 to 0.90 in 0.02 increments.

Rule

Choose the threshold that maximizes validation F1.

Final test

Apply the chosen threshold to 2023–2024 only; never tune on the test set.

12. Key Code Fix

Recorded error

The notebook failed on probability[1] because Spark represented probability as a vector UDT rather than a directly indexable array.

Fix

Convert the Spark vector to an array first, then extract element 1 as the positive-class probability.

Effect

Threshold selection and confusion-matrix calculations become compatible with the observed Spark environment.

13. Feature Importance

Tree models

Random Forest and GBT expose feature-importance values.

Mapping

Encoded positions are mapped back to readable numeric and categorical feature names.

Caution

Feature importance indicates predictive contribution in the fitted model and does not establish causality.

14. Sanity Checks

Early signal

Check correlation between early calendar-year citations and total citations.

Interpretation

Metadata-only is the primary research setting; metadata plus early citation tests whether near-term citation information improves prediction.

Reporting

Do not describe the secondary setting as metadata-only or as an exact first-12-month model.

15. Outputs

Results

Comparison table with feature mode, model, selected threshold, validation/test ROC-AUC and PR-AUC, precision, recall, F1, accuracy, and confusion-matrix counts.

Importance

CSV tables for Random Forest and GBT under both feature settings.

Thresholds

Validation threshold curves/tables for each model and setting.

Use

Outputs support the project report, figures, and results discussion.

16. Recommended Report Structure

Dataset

Source, retrieval filter, record counts, exclusions, final sample size, and class distribution.

Target

Year-specific top-10% definition and motivation.

Experiments

Separate metadata-only and metadata-plus-early-citation.

Models

Three classifiers and imbalance strategies.

Results

Emphasize PR-AUC, then ROC-AUC and positive-class metrics.

Validity

State that thresholds were selected using validation data only and final testing used 2023–2024.

17. Limitations

Citation window

OpenAlex calendar-year counts do not provide an exact first-12-month window.

Corpus

Results depend on the OpenAlex corpus and AI + climate retrieval definition.

Imbalance

The top-10% target creates a strongly imbalanced classification problem.

Generalization

Results on 2023–2024 do not guarantee identical performance on every future year/domain.

Causality

Predictive associations should not be interpreted as causal effects.

18. Practical Execution Checklist

Step 1

Install compatible Python, Java, PySpark, pandas, NumPy, matplotlib, and scikit-learn.

Step 2

Run setup/imports before downstream cells.

Step 3

Collect the OpenAlex corpus once or reuse the saved CSV.

Step 4

Inspect and clean the paper-level dataset.

Step 5

Construct yearly top-10% labels and inspect class counts.

Step 6

Build both feature settings.

Step 7

Train Logistic Regression, Random Forest, and GBT.

Step 8

Select the F1-maximizing threshold using 2022 validation.

Step 9

Evaluate once on 2023–2024.

Step 10

Compare metrics and inspect feature importance.

Step 11

Export CSV tables/figures and record sample sizes, class balance, parameters, thresholds, and final metrics.

19. Methodology Flow

OpenAlex AI + climate corpus → Cleaning → Year-specific top-10% label → Feature engineering → 2020–21 Train / 2022 Validation / 2023–24 Test → LR + RF + GBT → Validation threshold tuning → Final test metrics → Feature importance → Paper-ready outputs

20. Final Methodology Summary

This is a temporally ordered, leakage-aware binary classification study. The primary experiment uses publication metadata to predict membership in the top 10% of citations within each publication-year cohort. A secondary experiment adds an early calendar-year citation signal. The final evaluation uses the untouched 2023–2024 period, with PR-AUC emphasized because the positive class is strongly imbalanced.

Source Basis

This document is based on the submitted project presentation and supplied notebook. It preserves the stated OpenAlex corpus, yearly top-10% target, 2020–2021 training, 2022 validation, 2023–2024 test design, three classifiers, imbalance handling, evaluation metrics, validation-only threshold selection, feature-importance analysis, and the limitation concerning exact first-12-month citation measurement.